Design proposal for TE silicon waveguide group index extraction

Design objectives

I propose a set of unbalanced Mach–Zehnder interferometers (MZIs) to extract the group index of a silicon strip waveguide under quasi-TE polarization. Three primary devices have nominal arm length differences of 100 µm, 200 µm and 350 µm. Comparing their free spectral ranges (FSRs) will test the expected inverse relationship between FSR and arm imbalance and provide independent estimates of the same waveguide group index.

Two additional MZIs, each with a nominal 100 µm imbalance, compare Y-branch and multimode interference (MMI) connectors. A separate reference circuit connects two fibre grating couplers with a waveguide to measure the combined insertion loss of the measurement path, couplers and reference waveguide. The design contains five interferometers and one calibration circuit.

Waveguide and circuit geometry

The routed waveguides are 500 nm wide silicon strip waveguides, as recorded in the GDS cell metadata. The course fabrication process fixes the silicon thickness at 220 nm [1]. The design uses TE grating couplers intended for operation near 1550 nm, and the routed waveguide metadata specifies a 5 µm bend radius. The final MODE simulation must use the cladding and layer stack specified by the fabrication process.

Layout cell	Nominal ΔL	Splitter and combiner	Purpose
MZI100	100 µm	Y branch and broadband directional coupler	Group index
MZI200	200 µm	Y branch and broadband directional coupler	Group index
MZI350	350 µm	Y branch and broadband directional coupler	Group index
MZI_Y	100 µm	Y branch and Y branch	Connector loss comparison
MZI_MMI	100 µm	2 × 2 MMI and 2 × 2 MMI	Connector loss comparison
Reference	Not applicable	Two grating couplers and a 223.710 µm waveguide	Insertion loss reference

Table 1. Device set extracted from EBeam_yihua.gds. The reference is instantiated in the top cell EBeam_design_yi. Its two grating couplers are represented by a two-element array.

For the connector comparison, the routed single-mode path lengths are approximately 499.608 µm and 599.608 µm in both devices, excluding the connector interiors. This matching reduces propagation-length bias. Their bend counts differ, so the measurement will initially compare complete circuits rather than isolate connector excess loss.

Input port directory

Use the following exact GDS optical input labels to select the circuits. These are the opt_in measurement labels, rather than the internal component pin names such as opt1 or pin1.

Circuit	Exact optical input label
MZI100	opt_in_TE_1550_device_MZI100
MZI200	opt_in_TE_1550_device_MZI200
MZI350	opt_in_TE_1550_device_MZI350
Y connector MZI	opt_in_TE_1550_device_Ycon
MMI connector MZI	opt_in_TE_1550_device_MMI
Insertion loss reference	opt_in_TE_1550_device_insertloss

Table 2. Input names transcribed directly from the supplied GDS. Preserve capitalization and underscores when searching or assigning measurements.

Layout overview

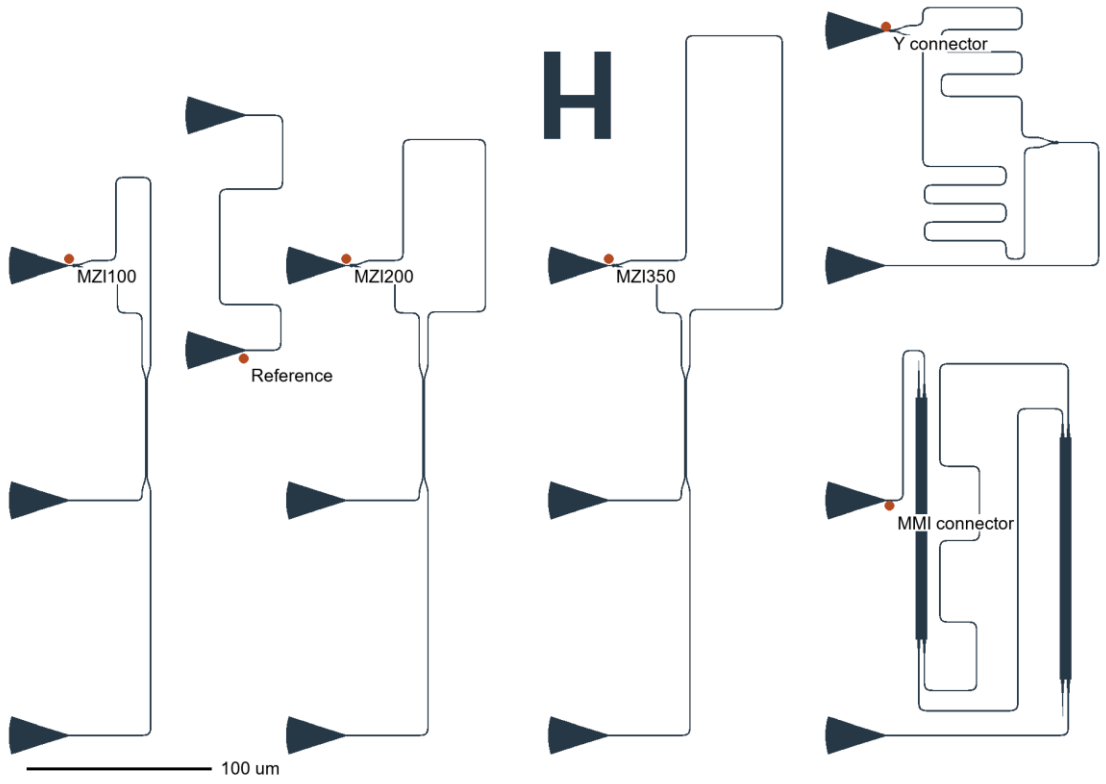


Figure 1. Silicon layer geometry rendered from the supplied GDS. Orange dots show the positions of the six optical input labels; they are not fibre alignment coordinates. The layout spans approximately 605 µm by 410 µm.

The three primary MZIs each provide two output grating couplers. Measuring both outputs provides a useful check on fringe periodicity and wavelength-dependent output imbalance. The connector comparison cells each have one measured output; unused MMI connections are terminated at the parent-cell level.